

MTH 446: Midterm # 1 Study

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Lecture Span

1. Study for midterm 1

Q1: General information about complex numbers

1. $|z|^2 = z \cdot \bar{z}$
2. Normalize z to find argument
3. $\cos \theta$ and $\sin \theta$ values at various θ 's

Q2: Roots of complex numbers

Remember to divide $2\pi k$ by n , where n is the root of the problem. Then, iterate through k . Also, $k = 0$ is the principal root.

Q3: Limits

1. Algebraic properties of limits and cont. functions
2. All polynomials are continuous
3. ϵ - δ definition

Q4: Demoire's formula

$$(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta \quad (1)$$

Q5: Sets

1. Boundaries (for all $\epsilon > 0$, $D_\epsilon(z_0)$ intersects with interior A and exterior $\mathbb{C}/A$)
2. Interior, $D_\epsilon(z_0) \subset A$ for all ϵ .
3. Exterior, the same but $\subset \mathbb{C}/A$
4. Open sets don't contain their boundary, closed is the contrary.
5. Closure
6. Accumulation points, deleted ϵ neighborhood.